

NITIN M

FULL-STACK DEVELOPER · AI/ML ENGINEER · GAME DEVELOPER

Bengaluru, India · nitinshakthi7@gmail.com · +91 9591746955 · [LinkedIn](#) · github.com/Nitinshakthi7

1 · ABOUT ME

Curious, highly inquisitive, and hands-on. I naturally want to understand how things work beneath the surface. I am not satisfied with simply knowing *what* something does — I want to know *why* it works, *how* it works, and whether there is a better way to build it.

I learn by **questioning, experimenting, breaking things, rebuilding them, and turning ideas into working projects.** When something interests me, I can become deeply involved in it, often going beyond what is required simply because I want to understand the subject properly. This has led me to explore full-stack development, AI/ML, cybersecurity, data systems, software architecture, and game development.

I am naturally a **problem solver and systems thinker.** I enjoy taking a complicated problem, breaking it into smaller pieces, understanding how those pieces interact, and then building a solution from the ground up. I also enjoy connecting ideas from different fields and using whatever technology is appropriate rather than restricting myself to one stack.

Creativity is another important part of how I work. I enjoy **building things that are not just functional but interesting** — from environmental intelligence and AI scheduling platforms to cybersecurity systems and a complete narrative game with its own mechanics, story and systems.

I value **independence and ownership.** I like figuring things out for myself, taking responsibility for the systems I build, documenting what I learn, and continuously improving my work. At the same time, I enjoy collaborative environments where people can challenge ideas, share knowledge, and build something better together.

Most importantly, I am **driven by curiosity rather than a checklist of technologies.** I want to become the kind of engineer who can face an unfamiliar problem, learn what is necessary, design a solution, and actually build it. I build with AI-assisted development as a core part of my workflow — writing specs, orchestrating coding agents and reviewing what they produce.

In short: I am a builder, a question-asker, a problem solver, and someone who genuinely enjoys learning by creating.

2 · SKILLS & EXPERIENCE

PROGRAMMING	Python · JavaScript · TypeScript · C++ · SQL · HTML · CSS
FRONTEND	React · Next.js · Vite · Tailwind CSS · Framer Motion · Recharts · shadcn/ui
BACKEND	Node.js · Express.js · Flask · REST APIs · JWT · Passport.js · Google OAuth 2.0
AI / ML	Scikit-Learn · XGBoost · Random Forest · SMOTE · PCA · K-Means · DBSCAN · Isolation Forest · GridSearchCV
DATA / DB	MongoDB · Mongoose · Prisma · SQLite · Redis
GAME DEV	Unreal Engine 5 · C++ · Blueprints · Pygame · Physics
TOOLS	Git · GitHub · Postman · Blender · VS Code · Vercel · Cron Jobs

EcoSense India — Environmental Intelligence Platform

[React](#) · [Node.js](#) · [Express](#) · [MongoDB](#) · [MapTiler](#) · [Recharts](#)

- Built a full-stack platform combining weather, AQI and satellite information from 10 data sources (ground stations, forecast models, satellites and reanalysis).
- Architected Express/MongoDB backend services and automated hourly data-collection, forecast and backfill pipelines.
- Added Google OAuth 2.0, GIS map layers, AI-powered insights and multi-city dashboards across 12h/24h/7d/30d trends.

IR-AIS — India Road Accident Intelligence System

[Python](#) · [XGBoost](#) · [Scikit-Learn](#) · [Next.js](#) · [TypeScript](#) · [SMOTE](#)

- Built severity classification for Slight, Serious and Fatal accidents from 12,316 records and engineered 26 predictive features.
- Used SMOTE to address class imbalance; implemented PCA/K-Means risk segmentation.
- Created an interactive Next.js predictor and model-comparison dashboard.

Behavioral Threat Detection Framework

[Python](#) · [Scikit-Learn](#) · [DBSCAN](#) · [Isolation Forest](#) · [Apriori](#)

- Analyzed 175K+ UNSW-NB15 network sessions using multiple unsupervised detection methods.
- Built association-rule mining and anomaly-consensus scoring to surface higher-confidence behavioral threats.

GameVault — Full-Stack Game Library

[Node.js](#) · [Express](#) · [MongoDB](#) · [JWT](#) · [REST APIs](#)

- Built 15+ RESTful endpoints using MVC architecture with JWT authentication.
- Implemented collections, wishlists, filtering, bulk insertion and cascade deletion.

Smart Energy Usage Tracker

[Node.js](#) · [Express](#) · [MongoDB](#) · [JWT](#)

- Designed Home → Room → Device → Reading data hierarchy with authenticated REST APIs.
- Implemented validation, energy-reading submission and cost analytics.

Used Car Price & Insurance Prediction

[Python](#) · [Scikit-Learn](#) · [Pandas](#) · [GridSearchCV](#)

- Built an ML pipeline over 2,805 records; achieved $R^2 = 0.883$ for price prediction and 81% insurance-status accuracy.
- Compared six supervised-learning algorithms and analyzed feature importance.

TimeGenius.AI — CICADA Hackathon

[Flask](#) · [Google OR-Tools](#) · [React](#) · [SQLite](#)

- Built conflict-free timetable generation for 50+ courses in under 3 seconds.
- Optimized faculty, room, workload, course-hour and availability constraints; added PDF/Excel export.

KOSMOS — Parametric Room & Asset Design Platform

[React](#) · [TypeScript](#) · [Vite](#) · [Tailwind CSS](#)

- Built a 2D design canvas with drag-and-drop furniture, parametric assets, JSON export and legacy migration.

Looped Lies — UE5 Narrative Action-Adventure

[Unreal Engine 5](#) · [C++](#) · [Blueprints](#) · [Nanite](#) · [Lumen](#)

- Developing a solo 15–20 hour narrative game with a 16,500-word story, five acts, three endings and a 742-iteration time loop.
- Building stealth, branching dialogue, moral alignment and save/load systems; authored a 1,341-line technical development guide.

AlgoViz — Interactive DSA Visualizer

[JavaScript](#) · [HTML](#) · [CSS](#) · [SVG](#)

- Built a framework-free, single-file learning tool covering 59 algorithms and data-structure operations across 8 categories.
- Added a step/play/rewind player with the executing code line, variables and pointers updating alongside each animation.

Don't Even Bother — Precision Platformer (Live on [itch.io](#))

[Python](#) · [Pygame](#) · [NumPy](#) · [OOP](#)

- Built a 19-level troll platformer across three acts, with a Brutal difficulty mode and a solver-bot "Race the Bot" feature.
- Synthesised every graphic, sound effect and music track at runtime — the game ships no art or audio files.

MerchantOS — Revenue Recovery Agent (Razorpay Buildathon)

[Next.js](#) · [React](#) · [TypeScript](#) · [Prisma](#) · [SQLite](#) · [Tailwind CSS](#)

- Built a five-stage agent loop (Detect → Diagnose → Decide → Act → Verify) that recovers failed payments, with every decision policy-bounded and audited.
- Ran a 50-case simulated batch end to end: 62.9% recovery rate, with opted-out customers correctly withheld.

MysteryDesk — Detective Investigation Game (Team of 4)

[React](#) · [Vite](#) · [Node.js](#) · [Express](#) · [SQLite](#)

- Rebuilt a teammate's single-case prototype into a full five-case detective game as part of a four-person team.
- Built a server-side contradiction engine comparing suspect testimony against evidence at query time, and five endings derived from the player's actual investigation.

Station Quest — Text Adventure

[React](#) · [TypeScript](#) · [Vite](#) · [Figma](#) · [Make](#)

- Built a text-based escape room with a natural-language command parser covering 16 verb groups and about 58 item aliases.
- Architected the parser, engine and world as separate modules so the parser could later be swapped for an LLM.

Snake & Ladder Quiz Game (Course Final Project)

[JavaScript](#) · [HTML](#) · [CSS](#)

- Built a two-player browser board game where trivia questions (Open Trivia DB) gate ladder climbs and snake dodges.
- Used the Random.org API for true-random dice rolls instead of Math.random.

Kanban Board (Course Project)

[Next.js](#) · [React](#) · [Prisma](#) · [NextAuth.js](#)

- Built a full-featured Kanban board entirely from prompts: drag-and-drop cards, four access roles, and a 10+ event activity log.
- Implemented credential authentication, rate limiting and an 8-model Prisma schema.

3 · CAREER DIRECTION

What I want: opportunities where I can work on real products, difficult engineering problems and technology with practical impact. I am especially interested in full-stack product engineering, AI/ML applications, backend systems, data-driven products, automation and game technology.

Where I want to grow: teams that value curiosity, experimentation, engineering depth and ownership — where I can learn from strong engineers while taking responsibility for meaningful parts of a product.

Companies I aspire to work toward: Google · Microsoft · NVIDIA · OpenAI · Amazon · Apple · Adobe · Epic Games · Unity and other high-impact technology/product companies.

Long-term direction: become an engineer who can take a problem from **idea** → **architecture** → **implementation** → **intelligent system** → **finished product**, with deep expertise in software engineering and AI.

4 · EDUCATION

Atria University — Bengaluru

2024 – 2028 (Expected)

B.Tech in Digital Transformation · Major: Digital Transformation · Minor: Artificial Intelligence & Machine Learning

New Horizon Pre-University College — Bengaluru

2024

PUC / 12th Grade — 60%

SJR Public School — Bengaluru

2022

10th Grade — 65%

SELECTED STRENGTHS

Systems Thinking · Problem Solving · Rapid Prototyping · Technical Documentation · Creative Thinking · Collaborative Development · Learning Through Building